Multivariate and Velocity-Resolved Reverberation in NGC 5548: Astrophysical Motivation for an Information-Based Analysis of Continuum–BLR Coupling

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ABSTRACT

Reverberation mapping constrains the size, geometry, and mass of supermassive black holes in active galactic nuclei (AGN) by measuring the delay between continuum and emission-line variability. Pairwise linear methods like the Interpolated Cross-Correlation Function (ICCF) or transfer function modeling (JAVELIN) assume a direct, linear mapping between a single driver and line response. In this work, we present a multivariate, velocity-resolved analysis of NGC 5548 using the Synergistic–Unique–Redundant Decomposition (SURD) information-theoretic framework to identify multivariate predictive-information structures between the 5100 Å optical continuum and the Hβ line wings and core. Enforcing a strict-overlap preprocessing window (MJD 47512–49255), we find that classical linear methods identify prompt light-travel delays of 13–20 days: ICCF peaks at 13.0 d (Red), 19.0 d (Blue), and 20.0 d (Core), matching JAVELIN's lag posteriors (12.8 ± 1.8 d Red, $19.4^{+2.5}_{-2.9}$ d Blue, and $20.1^{+2.9}_{-3.2}$ d Core). In contrast, SURD synergy scans peak at much longer, scan-range-dependent timescales (56 d Red, 109 d Blue, and 159 d Core). By running a synthetic negative control test with zero true long-lag coupling, we demonstrate that these unconditioned long-lag peaks are spurious artifacts manufactured by seasonal observing gaps, interpolation-induced red-noise coherence, and normalized synergy inflation as the denominator $I(Y; X_1, X_2)$ approaches zero. Furthermore, target-history conditioning (conditional PID) shifts the synergy peak to 73 d, but surrogate testing reveals that this peak is statistically insignificant (global max-statistic $p \approx 0.98$) due to finite-sample bias in the extremely sparse 4D joint probability cells (where 89.2% of bins are completely empty). This frames our study as a cautionary, methodological case study showing that multivariate information scans are highly susceptible to windowing, boundary, and finite-sample artifacts, highlighting the absolute necessity of rigorous global multiple-testing controls and negative controls in time-domain astronomy.

Key words: galaxies: active – galaxies: nuclei – galaxies: individual: NGC 5548 – methods: statistical – methods: observational

1 INTRODUCTION

Active galactic nuclei (AGN) are powered by accretion onto supermassive black holes and radiate over a broad range of wavelengths through a compact but highly structured central engine. The standard observational picture includes an accretion disk that produces much of the optical and ultraviolet continuum, a hot corona responsible for the X-ray emission, a broad-line region (BLR) composed of dense photoionized gas moving at velocities of order 10^3 to 10^4 km s^{-1}, and more distant dusty material that reprocesses the radiation field at infrared wavelengths Peterson (2004); Netzer (2013); Cackett et al. (2021). Because these regions are too compact to resolve directly in most AGN, temporal variability provides one of the few effective ways to infer their sizes, structure, and coupling.

Reverberation mapping exploits the fact that fluctuations in the central ionizing source are echoed by delayed responses in remote reprocessing or emission-line regions. In its simplest linearized form, the response of an emission line $L(t)$ to a driving continuum $C(t)$ may be written as

$$L(t) = \int \Psi(\tau)\, C(t-\tau)\, d\tau, \tag{1}$$

where $\Psi(\tau)$ is the transfer function or delay distribution. The classical use of this formalism has been to derive a characteristic lag τ, associate it with a BLR radius $R \sim c\tau$, and combine that scale with the line width to estimate the black-hole mass under a virial assumption Peterson (2004); Bentz & Katz (2015). This framework has been extremely successful and remains foundational to AGN black-hole mass scaling relations.

Nevertheless, the astrophysical interpretation of reverberation data has become progressively more subtle. The quantity that drives the broad emission lines is the ionizing continuum, particularly in the extreme ultraviolet and soft X-ray bands, whereas the continuum actually monitored in many optical reverberation campaigns is a more accessible proxy, often the flux near 5100 Å. As a result, the measured lag between an observed continuum band and an emission line need not correspond to a direct causal connection between those two observables alone. Instead, both may be responding, with different delays and different filtering, to a latent radiation field that is only

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imperfectly traced by the observed bands Cackett et al. (2021). The question is therefore not merely what the lag is, but whether the monitored continuum contains the relevant information needed to explain the line response.

This issue is underscored by continuum reverberation mapping. A series of high-cadence monitoring programs has revealed that ultraviolet and optical continuum bands in AGN are delayed with respect to one another, with delays that often grow systematically with wavelength as expected for a thermally stratified accretion disk. However, the measured lags are frequently larger than predicted by the simplest geometrically thin, optically thick accretion-disk models Fausnaugh et al. (2016); Fausnaugh (2017); Edelson et al. (2019); Cackett et al. (2021). These results suggest that the observed continuum variability is not fully described by a minimal reprocessing picture and may involve additional processes such as intrinsic accretion-rate fluctuations, diffuse continuum emission, complex irradiation geometry, or nontrivial radiative transfer. From the standpoint of BLR reverberation, this implies that optical continuum light curves may not be interpreted as pristine drivers without further caution.

NGC 5548 is the canonical source in which these concerns become impossible to ignore. It is one of the most intensively monitored Seyfert galaxies and has served as a cornerstone of reverberation mapping for decades. It has played a major role in calibrating broad-line lags, virial mass estimates, and radius–luminosity relations, while more recent campaigns have transformed it into a benchmark for multiwavelength and velocity-resolved reverberation analysis Peterson et al. (2002); Bentz & Katz (2015); Cackett et al. (2021). In particular, the AGN STORM campaign delivered one of the richest reverberation datasets ever obtained, including optical and ultraviolet continuum monitoring, extensive spectroscopy, and detailed velocity-delay information for several broad lines De Rosa et al. (2015); Pei et al. (2017); Horne et al. (2021).

These observations revealed that NGC 5548 is not well described by a stationary one-zone BLR driven by a single observed continuum proxy. Velocity-resolved reverberation results showed clear structure across the Hβ profile, with the line wings and core participating differently in the reverberation pattern Pei et al. (2017); Horne et al. (2021). The interpretation of these data points toward a BLR that is radially stratified and kinematically complex, possibly including multiple components or zones rather than a single homogeneous distribution of clouds Horne et al. (2021); Long & Dexter (2025). This immediately motivates a shift away from purely integrated-line analyses. If the blue wing, line core, and red wing carry different dynamical information, then treating Hβ as a single scalar time series necessarily compresses part of the astrophysical content of the data.

An even more striking complication emerged during the same source's monitoring history: the so-called "holiday" in NGC 5548. During part of the AGN STORM campaign, the broad emission lines became partially decorrelated from the observed UV continuum, even though they had previously tracked it closely. The leading interpretation is that an obscuring wind or shielding structure modified the ionizing spectral energy distribution reaching the BLR while leaving the observed UV continuum comparatively unaffected Dehghanian et al. (2019). The importance of this result cannot be overstated. It demonstrates directly that an observed continuum channel may cease to be a faithful tracer of the radiation field relevant for BLR photoionization. Consequently, a line–continuum lag analysis may look physically incomplete not because the line has stopped reverberating, but because the monitored continuum is no longer the appropriate driver proxy.

This broader lesson is now reinforced by work across the reverberation-mapping literature. Velocity-resolved Hβ studies increasingly show that BLR kinematics are diverse across AGN and may evolve over time, with some sources displaying signatures consistent with inflow, others with outflow, and others with virialized or mixed dynamics Du et al. (2018); Wang et al. (2025); Feng et al. (2024). In parallel, changing-look AGN have demonstrated that the relation between continuum state and broad-line emission can undergo dramatic reorganization on humanly accessible timescales Ricci & Trakhtenbrot (2023). Although changing-look phenomena are not the focus of the present study, they help establish a wider astrophysical principle: the mapping between observed continuum, ionizing continuum, and line-emitting gas is not guaranteed to be stationary.

Within this context, NGC 5548 is particularly well suited for a more conservative, multivariate analysis of the reverberation problem. The source offers high-quality continuum monitoring, extensive Hβ spectroscopy, and a rich legacy of prior reverberation results against which new analyses may be compared. At the same time, it is sufficiently complex to require such an approach. If one asks only for an integrated Hβ lag relative to the optical continuum, much of the internal structure of the reverberation process is averaged away. By contrast, if one treats the blue wing, core, and red wing of Hβ as distinct observables, new astrophysical questions become accessible. Does the core behave primarily as an echo of the continuum, or does it also depend on information already present in the wings? Are the two wings symmetric in their predictive role, as one might expect in an idealized virialized BLR, or does one side of the profile carry systematically different dynamical information? Can the observed optical continuum alone account for the velocity-resolved response, or do hidden drivers and internal line-component couplings remain necessary?

These questions are not purely methodological. They arise naturally from long-standing astrophysical uncertainties about the BLR and the continuum source. The BLR is expected to be sensitive to the ionizing continuum, but in practice the ionizing continuum is usually unobserved. The optical continuum is measurable but is itself part of a reverberating accretion-flow system and may therefore contain both intrinsic and reprocessed variability components Fausnaugh (2017); Cackett et al. (2021). The Hβ line is observable with high fidelity, but its integrated flux merges together multiple kinematic regions whose relative responses may differ. Once velocity-resolved spectroscopy is available, the astrophysical challenge becomes one of causal disentanglement among observables that are neither independent nor trivially ordered in a simple driver–response chain.

A number of established approaches address parts of this challenge. Interpolated cross-correlation methods and related lag estimators remain the standard tools for continuum–line reverberation analysis and continue to provide essential baseline measurements Peterson (2004); White & Peterson (1994). JAVELIN and related stochastic-process approaches model reverberation as a damped random walk continuum filtered by a transfer function, thereby improving lag estimation and uncertainty quantification in many cases Zu et al. (2011, 2013). Velocity-delay mapping and dynamical modeling aim to reconstruct the geometry and kinematics of the BLR directly from the line-profile response Pancoast et al. (2014); Horne et al. (2021). Continuum reverberation studies focus on interband disk lags and the thermal structure of the accretion flow Fausnaugh et al. (2016); Fausnaugh (2017); Edelson et al. (2019). Each of these approaches has been productive, but they are typically tailored to a specific projection of the problem: pairwise lags, transfer-function fitting, dynamical inversion, or continuum size scaling.

What remains comparatively underexplored is the specific astrophysical question of how multiple simultaneously observed channels jointly constrain the future evolution of one chosen component of the system, especially when those channels include both the continuum and velocity-resolved line segments. In other words, once the Hβ profile has been decomposed into blue wing, core, and red wing light curves, and once a conservative common time interval is imposed, the natural problem is no longer simply "what is the lag?" It becomes: how much of the future behavior of one velocity component is already encoded in the continuum, how much is encoded in another part of the line profile, and how much lies beyond the information available in the monitored observables? That is the astrophysical problem motivating the present paper.

The emphasis on strict temporal overlap is also not incidental. Irregular cadence, seasonal gaps, and unequal overlap among observed channels are intrinsic features of AGN monitoring data. Any multivariate time-series analysis is therefore vulnerable to spurious structure introduced by aggressive interpolation, extrapolation across gaps, or implicit assumptions about the simultaneity of channels that were not truly observed together. In the context of NGC 5548, where the scientific questions depend sensitively on comparing the optical continuum to distinct Hβ velocity components, it is essential to work under conservative overlap conditions. The goal is not to maximize sample size at all costs, but to preserve physical interpretability. A shorter but genuinely shared time interval is preferable to a longer one in which apparent multivariate structure may be partly synthetic.

The present paper is therefore built around a deliberately astrophysical motivation. We seek to determine how the optical continuum near 5100 Å and the velocity-resolved components of Hβ in NGC 5548 are coupled when the data are restricted to robust common overlap. Our interest is not merely in recovering a known lag, but in characterizing the continuum–BLR system as a multichannel, time-dependent structure. We focus on the blue wing, line core, and red wing because these components are directly linked to the low-ionization BLR geometry and kinematics, and because they allow one to ask whether different parts of the line profile participate differently in the reverberation process. This is the astrophysical motivation for the analysis developed in the sections that follow.

The methodological framework used later in the paper is based on information decomposition, but we intentionally postpone its formal description. The present Introduction has a different role: to show why such an analysis is astrophysically warranted. The literature has already established that NGC 5548 exhibits continuum interband lags, velocity-resolved Hβ structure, partial line–continuum decoupling during the holiday, and evidence for a multicomponent BLR. The natural next step is not simply to add another lag estimator, but to investigate the reverberation data in a way that is explicitly multivariate, velocity-resolved, and conservative with respect to temporal overlap. That is the problem we address here.

The structure of this paper is as follows. In Section 2 we describe the observations, spectral processing, and the construction of strictly overlapping continuum and velocity-resolved Hβ time series. In Section 3 we introduce the formal information-based framework used to quantify the coupling among the observables. In Section 4 we present the main results for continuum–line and line–line relationships, including robustness tests and comparisons with classical reverberation measures. In Section 5 we discuss the implications for the structure of the BLR in NGC 5548, the role of hidden or imperfectly observed drivers, and the interpretation of multivariate reverberation signals. Section 6 summarizes our conclusions and outlines possible extensions to larger AGN samples and to other broad emission lines.

2 SYNERGISTIC–UNIQUE–REDUNDANT DECOMPOSITION (SURD) FOR TIME SERIES

2.1 Problem formulation

Consider a collection of N observed stochastic processes,

$$\mathbf{Q}(t) = \big(Q_1(t), Q_2(t), \ldots, Q_N(t)\big), \tag{2}$$

sampled at discrete times $t = t_n$. For a chosen target index j, the goal is to quantify how much information about the future state

$$Q_j^+ \equiv Q_j(t + \Delta T) \tag{3}$$

is contained in the observed variables at time t and, crucially, to decompose that predictive information into *redundant*, *unique*, and *synergistic* contributions Martínez-Sánchez et al. (2024).

The central motivation is that in multivariate systems, pairwise predictive measures such as Granger causality or transfer entropy can establish that a variable is informative about a target, but they do not in general distinguish whether that information is:

(i) unique to one predictor,
(ii) duplicated across several predictors, or
(iii) available only from a joint observation of several predictors.

SURD is designed precisely to perform that decomposition in a time-series setting Martínez-Sánchez et al. (2024); Schreiber (2000); Williams & Beer (2010).

2.2 Information-theoretic preliminaries

Let $Y \equiv Q_j^+$ denote the target and let $\mathbf{X}$ denote the vector of observed predictors. For discrete variables, the Shannon entropy is

$$H(Y) = -\sum_{y} p(y) \log_2 p(y), \tag{4}$$

and the conditional entropy of Y given $\mathbf{X}$ is

$$H(Y \mid \mathbf{X}) = -\sum_{y,\mathbf{x}} p(y,\mathbf{x}) \log_2 p(y \mid \mathbf{x}). \tag{5}$$

The mutual information between Y and $\mathbf{X}$ is

$$I(Y;\mathbf{X}) = H(Y) - H(Y \mid \mathbf{X}) = \sum_{y,\mathbf{x}} p(y,\mathbf{x}) \log_2 \frac{p(y,\mathbf{x})}{p(y)p(\mathbf{x})}. \tag{6}$$

This quantity measures the total predictive information about the future target that is available in the observed present state Shannon (1948); Cover & Thomas (2006).

In SURD, the unexplained part of the future is interpreted as an *information leak*,

$$\Delta I_{\mathrm{leak}\to j} = H(Y \mid \mathbf{X}), \tag{7}$$

so that the forward information balance reads

$$H(Y) = I(Y;\mathbf{X}) + \Delta I_{\mathrm{leak}\to j}. \tag{8}$$

Equation (8) is one of the core ideas of SURD: the future uncertainty of the target splits into an observed part, explained by the measured variables, and an unobserved part, attributed to hidden or unmeasured influences Martínez-Sánchez et al. (2024).

2.3 From total predictive information to information-decomposition components

Let the predictor vector be

$$\mathbf{X} = (X_1, \ldots, X_M), \tag{9}$$

where the X_α may correspond either to different physical variables at the same time, or to different lagged copies of the same variables. SURD decomposes the observable predictive information $I(Y;\mathbf{X})$ into additive components:

$$I(Y;\mathbf{X}) = \sum_{\mathcal{A}\in C} \Delta I^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \Delta I^S_{\mathcal{A}\to j}, \tag{10}$$

where:

- $\Delta I^R_{\mathcal{A}\to j}$ is the *redundant* contribution from the group of predictors indexed by $\mathcal{A}$,
- $\Delta I^U_{\alpha\to j}$ is the *unique* contribution of predictor X_α,
- $\Delta I^S_{\mathcal{A}\to j}$ is the *synergistic* contribution from the group $\mathcal{A}$,
- C denotes all non-singleton subsets of $\{1,\dots,M\}$.

The full future entropy therefore obeys

$$H(Y) = \sum_{\mathcal{A}\in C} \Delta I^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \Delta I^S_{\mathcal{A}\to j} + \Delta I_{\mathrm{leak}\to j}. \tag{11}$$

The three information-decomposition types have the following interpretation Martínez-Sánchez et al. (2024):

- Redundant: the same predictive information is shared across several predictors,

- Unique: a predictor contains information not available from any other predictor alone,

- Synergistic: information appears only when several predictors are observed jointly.

2.4 Specific mutual information and statewise decomposition

A key feature of SURD is that the source of predictability may depend on which *particular future event* y occurs. To resolve this dependence, SURD works first at the level of *specific mutual information* DeWeese & Meister (1999); Butts (2003); Martínez-Sánchez et al. (2024).

For a particular target state $Y = y$, the specific mutual information between y and a predictor set $\mathbf{X}_\mathcal{A}$ is

$$\tilde{\imath}_\mathcal{A}(y) \equiv \tilde{\imath}(y;\mathbf{X}_\mathcal{A}) = \sum_{\mathbf{x}_\mathcal{A}} \frac{p(y,\mathbf{x}_\mathcal{A})}{p(y)} \log_2 \frac{p(y,\mathbf{x}_\mathcal{A})}{p(y)\,p(\mathbf{x}_\mathcal{A})}. \tag{12}$$

The ordinary mutual information is recovered by averaging over target states:

$$I(Y;\mathbf{X}_\mathcal{A}) = \sum_y p(y)\,\tilde{\imath}_\mathcal{A}(y). \tag{13}$$

The conceptual reason for introducing $\tilde{\imath}_\mathcal{A}(y)$ is that different predictors may be informative for different target outcomes. A predictor may, for example, strongly constrain large positive target excursions while another may mainly constrain negative or low-amplitude states. A decomposition performed only at the level of the average mutual information can obscure this state dependence. SURD therefore computes the information contributions for each y, decomposes them into redundant, unique, and synergistic increments, and only afterwards averages over y Martínez-Sánchez et al. (2024).

2.5 Statewise incremental construction of SURD

Let $\mathfrak{P}^*(\{1,\dots,M\})$ denote the set of all non-empty subsets of predictors. For each subset $\mathcal{A} \in \mathfrak{P}^*(\{1,\dots,M\})$, one computes

$$\tilde{\imath}_\mathcal{A}(y) = \tilde{\imath}(y;\mathbf{X}_\mathcal{A}). \tag{14}$$

SURD then considers the collection of all these values for a fixed target state y and identifies the *increments* in predictive information that arise as one moves from smaller to larger subsets of predictors Martínez-Sánchez et al. (2024). Denoting these statewise increments schematically by

$$\Delta\tilde{\imath}_\mathcal{A}(y), \tag{15}$$

the classification rule is:

- an increment assigned to one predictor alone contributes to $\Delta\tilde{\imath}^U_\alpha(y)$,
- an increment shared across several predictors contributes to $\Delta\tilde{\imath}^R_\mathcal{A}(y)$,
- an increment that arises only when a collection of predictors is observed jointly contributes to $\Delta\tilde{\imath}^S_\mathcal{A}(y)$.

The averaged decomposed components are then

$$\Delta I^R_{\mathcal{A}\to j} = \sum_y p(y)\,\Delta\tilde{\imath}^R_\mathcal{A}(y), \tag{16}$$

$$\Delta I^U_{\alpha\to j} = \sum_y p(y)\,\Delta\tilde{\imath}^U_\alpha(y), \tag{17}$$

$$\Delta I^S_{\mathcal{A}\to j} = \sum_y p(y)\,\Delta\tilde{\imath}^S_\mathcal{A}(y). \tag{18}$$

This construction ensures that the total observable predictive information is recovered exactly:

$$I(Y;\mathbf{X}) = \sum_{\mathcal{A}\in C} \Delta I^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \Delta I^S_{\mathcal{A}\to j}. \tag{19}$$

2.6 Two-source case

For two predictors X_1 and X_2, the decomposition reduces to

$$I(Y;X_1,X_2) = \Delta I^R_{12\to j} + \Delta I^U_{1\to j} + \Delta I^U_{2\to j} + \Delta I^S_{12\to j}. \tag{20}$$

Consistency with the single-predictor mutual informations requires

$$I(Y;X_1) = \Delta I^R_{12\to j} + \Delta I^U_{1\to j}, \tag{21}$$

$$I(Y;X_2) = \Delta I^R_{12\to j} + \Delta I^U_{2\to j}. \tag{22}$$

Hence the synergistic contribution may be written as the remainder

$$\Delta I^S_{12\to j} = I(Y;X_1,X_2) - \Delta I^R_{12\to j} - \Delta I^U_{1\to j} - \Delta I^U_{2\to j}. \tag{23}$$

This form makes the interpretation transparent: the synergy is the part of the joint predictive information that cannot be accounted for by the redundant and unique terms.

2.7 Extension to time series with multiple lags

For time-series applications, the predictor vector is typically augmented with lagged copies of the observed variables. If one includes p past lags per variable, the predictor vector becomes

$$\begin{aligned}\mathbf{X}(t) = \Big[&Q_1(t), Q_1(t-\Delta T_1), \dots, Q_1(t-\Delta T_p),\\ &Q_2(t), Q_2(t-\Delta T_1), \dots, Q_N(t-\Delta T_p)\Big].\end{aligned} \tag{24}$$

For example, with $N = 2$ variables and one additional lag,

$$\mathbf{X}(t) = \big[Q_1(t), Q_1(t-\Delta T), Q_2(t), Q_2(t-\Delta T)\big]. \tag{25}$$

SURD then treats each lagged component as one predictor channel and applies the same decomposition machinery to the enlarged predictor set Martínez-Sánchez et al. (2024). In this way, the method can distinguish:

- self-causation from $Q_j(t-\Delta T_k)$ to $Q_j(t+\Delta T)$,
- cross-causation from $Q_i(t-\Delta T_k)$ to $Q_j(t+\Delta T)$,
- redundancy between different lags,
- synergy between variables and lags.

This is particularly important when several lagged channels may encode overlapping predictive content. For example, adding $Q_j(t)$ and $Q_j(t-\Delta T)$ can increase redundancy rather than revealing new independent predictive pathways. SURD is expressly designed to make that distinction Martínez-Sánchez et al. (2024).

2.8 Contemporaneous and mixed lagged dependencies

A further advantage of the formalism is that the predictor vector can include variables at time $t+\Delta T$ other than the target itself, thereby allowing the analysis of contemporaneous or effectively sub-resolution interactions Martínez-Sánchez et al. (2024). In practice, one excludes the target variable from the contemporaneous predictor set to avoid trivial self-determination, but permits other variables measured on the same time slice. Thus, a generic predictor vector may mix present, lagged, and contemporaneous channels:

$$\mathbf{X}(t) = \big[\text{past lags, present variables,} \\ \text{contemporaneous non-target variables}\big]. \tag{26}$$

The same decomposition (11) then applies.

2.9 Normalization and interpretation

Because the observable predictive information is conserved by construction, the unique, redundant, and synergistic terms are naturally normalized by the total observable mutual information:

$$\widehat{\Delta I}^R_{\mathcal{A}\to j} = \frac{\Delta I^R_{\mathcal{A}\to j}}{I(Y;\mathbf{X})}, \qquad \widehat{\Delta I}^U_{\alpha\to j} = \frac{\Delta I^U_{\alpha\to j}}{I(Y;\mathbf{X})}, \qquad \widehat{\Delta I}^S_{\mathcal{A}\to j} = \frac{\Delta I^S_{\mathcal{A}\to j}}{I(Y;\mathbf{X})}. \tag{27}$$

These normalized terms sum to unity:

$$\sum_{\mathcal{A}\in C} \widehat{\Delta I}^R_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \widehat{\Delta I}^U_{\alpha\to j} + \sum_{\mathcal{A}\in C} \widehat{\Delta I}^S_{\mathcal{A}\to j} = 1. \tag{28}$$

The leak is normalized instead by the total entropy of the target:

$$\widehat{\Delta I}_{\text{leak}\to j} = \frac{\Delta I_{\text{leak}\to j}}{H(Y)} = \frac{H(Y \mid \mathbf{X})}{H(Y)}. \tag{29}$$

Hence:

- $\widehat{\Delta I}_{\text{leak}\to j} \approx 0$ means the observed predictors account for almost all predictive information about the target,
- $\widehat{\Delta I}_{\text{leak}\to j} \approx 1$ means most of the target's future uncertainty remains unexplained by the observed variables.

2.10 Practical estimation

In applications, the probabilities appearing in Eqs. (6) and (12) must be estimated from finite data. The original SURD work uses histogram-based and transport-map-based estimators depending on dimensionality and sample size Martínez-Sánchez et al. (2024). In a histogram implementation, one discretizes the predictor-target space into bins and estimates

$$p(y,\mathbf{x}) \approx \frac{n(y,\mathbf{x})}{N_{\text{samples}}}, \tag{30}$$

where $n(y,\mathbf{x})$ is the bin count. The resulting empirical joint distribution is then used to compute all required marginal and conditional probabilities. More sophisticated estimators, such as transport maps, are advantageous in high dimensions because the number of bins required by direct histograms grows exponentially with the number of predictors.

Algorithmically, the SURD workflow for a target Q_j^+ is:

(i) choose the predictor vector $\mathbf{X}$, including variables, lags, or contemporaneous channels;
(ii) estimate the probability distributions $p(y,\mathbf{x}_{\mathcal{A}})$ for all relevant predictor subsets $\mathcal{A}$;
(iii) compute the specific informations $\tilde{\imath}_{\mathcal{A}}(y)$ for every target state y;
(iv) decompose the statewise increments into redundant, unique, and synergistic parts;
(v) average over y to obtain ΔI^R, ΔI^U, and ΔI^S;
(vi) compute the leak from Eq. (7);
(vii) normalize the observable components by $I(Y;\mathbf{X})$ and the leak by $H(Y)$.

2.11 Properties and assumptions

SURD has several properties that are important in time-series applications Martínez-Sánchez et al. (2024):

(i) **Non-negativity:** all redundant, unique, synergistic, and leak terms are non-negative.
(ii) **Additivity:** the observable predictive information is exactly partitioned into R, U, and S terms.
(iii) **Sensitivity to nonlinear dependence:** because the construction is based on mutual information, it is not restricted to linear couplings.
(iv) **Compatibility with lagged and contemporaneous predictors:** time-lag structure enters through the predictor vector.
(v) **Hidden-variable diagnostics:** the leak quantifies the fraction of future uncertainty left unexplained by the observed variables.

The main assumptions are those common to observational information-theoretic analysis:

- the processes are sufficiently stationary over the analysis interval for probability estimation to be meaningful,
- the chosen time resolution is physically relevant,
- the sample size is adequate for the selected probability estimator,
- and statistical interpretation remains limited to the observed variables plus the quantified unresolved contribution represented by the leak.

2.12 Summary

SURD starts from the predictive information $I(Q_j^+;\mathbf{X})$ between a future target and a multivariate predictor state. It then decomposes that predictive information into state-averaged unique, redundant, and synergistic contributions, while retaining the conditional entropy $H(Q_j^+ \mid \mathbf{X})$ as an explicit measure of hidden or unobserved influence. Its time-series extension is straightforward: past lags, present variables, and even contemporaneous non-target channels are simply incorporated into the predictor vector. For multivariate dynamical systems, this yields a decomposition that is richer than pairwise lag measures because it distinguishes direct predictive effects from duplicated information and from genuinely joint multivariate effects Martínez-Sánchez et al. (2024); Williams & Beer (2010).

3 OBSERVATIONS, SPECTRAL PROCESSING, AND STRICT OVERLAP ALIGNMENT

To evaluate the capabilities of the SURD algorithm on real-world astrophysical data, we analyze the well-monitored Seyfert 1 galaxy NGC 5548. This source has a rich legacy of reverberation mapping, including the high-cadence AGN STORM campaign De Rosa et al. (2015); Pei et al. (2017). Our analysis utilizes two main datasets: the optical continuum flux at 5100 Å and velocity-resolved Hβ emission line profiles.

3.1 Data Products and MJD Correction

The 5100 Å continuum light curve is obtained from the AGN Watch databases (specifically, `c5100.dat`), spanning a long duration of several years. The integrated Hβ emission line light curve is similarly taken from clean AGN Watch compilations (`ngc5548_agnwatch_hb_clean.csv`).

The velocity-resolved spectroscopy files, containing Hβ profiles, were processed from individual `.spc` files. The filenames (following the template `nXXXXXma.spc`) contain an embedded time indicator where the numerical part `XXXXX` represents a Julian Date (JD) offset. Specifically, we confirmed the filename-to-MJD conversion:

$$\mathrm{MJD} = \mathrm{XXXXX} - 10000. \tag{31}$$

Applying this correction aligns the spectroscopic observations from the filename values in the range 57512–59255 to the physical Modified Julian Date (MJD) range of 47512–49255, matching the historical campaigns of the late 1980s and early 1990s.

We extract three kinematic components from the Hβ profile:

(i) **Blue Wing:** integrated flux in the velocity range $[-3000, -1000]\ \mathrm{km\,s^{-1}}$.
(ii) **Line Core:** integrated flux in the velocity range $[-1000, +1000]\ \mathrm{km\,s^{-1}}$.
(iii) **Red Wing:** integrated flux in the velocity range $[+1000, +3000]\ \mathrm{km\,s^{-1}}$.

A summary of the raw data products is presented in Table 1.

3.2 Strict Overlap Interpolation and Cadence Homogenization

Because the velocity-resolved spectral campaign is restricted to the interval MJD 47512 to 49255, any multivariate analysis must be confined strictly to this common window. In previous exploratory versions of the analysis (e.g., V10), the full range of the continuum was processed, leading to edge effects and spurious correlations from forward/backward filling of missing line profile data.

In this work (V11 Strict Overlap), we enforce a strict temporal boundary:

$$t \in [\max(t_{\min}), \min(t_{\max})] = [47512.00, 49255.00]\ \mathrm{MJD}. \tag{32}$$

Within this window, we resample the light curves onto a uniform 1.0-day grid. The fluxes are interpolated using 1D linear interpolation, and any epoch outside the strictly observed boundaries is discarded, resulting in $N = 1744$ clean, synchronized epochs. Prior to running the SURD algorithm, all light curves are normalized using the z-score:

$$Z(t) = \frac{x(t) - \mu_x}{\sigma_x}, \tag{33}$$

where μ_x and σ_x are the mean and standard deviation over the synchronized overlap window. The resulting aligned and standardized light curves are visualized in Figure 1.

4 INFORMATION DECOMPOSITION AND LAG ANALYSIS RESULTS

We run the SURD algorithm on the prepared NGC 5548 data to study the predictive information flow between the continuum (S_1), the wings, and the core of Hβ.

4.1 Dominant Lags and Extended Scan Range

To test the physical stability of the synergy and leak profiles, we extended the lag search range from the initial limit of 60 days to 200 days. The comparison of peak lags and synergy values between the standard and extended scans is summarized in Table 2, and the complete information decomposition profiles with Monte Carlo uncertainty bands are displayed in Figure 2.

For all three velocity components, the peak synergy lag shifts when the scan range is extended from 60 days to 200 days. Specifically, the Core peak shifts from 45 days to 159 days, the Red Wing peak shifts from 39 days to 56 days, and the Blue Wing peak shifts from 53 days to 109 days. This extreme variability shows that the unconditioned peaks are highly scan-range-dependent and do not converge on stable physical delays, suggesting that they are boundary or windowing artifacts rather than real delays.

4.2 Surrogate and Null Hypothesis Testing

To check the statistical properties of the synergy scans, we perform two distinct classes of null tests:

(i) **Circular Shift Surrogates:** The predictor signals are circularly shifted relative to one another. This destroys the physical phase alignment but preserves the individual signal amplitudes and autocorrelation.
(ii) **Block Bootstrap Surrogates:** To preserve the short-term red-noise properties and autocorrelation of the AGN light curves, we apply block shuffling with a block size of 10.0 days.

At the peak synergy lags identified in the extended scan, the real unnormalized synergy values are 0.0526 bits for the Core (at 159 days), 0.5677 bits for the Red Wing (at 56 days), and 0.2198 bits for the Blue Wing (at 109 days). For the standard lag range (1–60 days), the real unnormalized synergy curves for all three targets locally exceed the 95th percentile surrogate null envelopes (for example, the Core unnormalized synergy at the standard 45-day peak is 0.1883 bits, which is well above the circular shift threshold of ~ 0.136 bits and the block bootstrap threshold of ~ 0.086 bits). However, in the extended scan (up to 200 days), while the Red and Blue wing unnormalized peaks exceed the local surrogate null envelopes, the Core peak at 159 days does not.

Crucially, after applying the Benjamini–Hochberg False Discovery Rate (FDR) correction for multiple testing across the 200 scanned lags, **none of the individual lags remain globally statistically significant for any of the three targets.**

This negative FDR result has an important methodological consequence: the observed synergy peaks should not be interpreted as mathematically precise delta-function physical delays (like a simple light-travel time). Instead, they represent broad, lag-dependent coupling zones where information flow locally exceeds noise, but scanning over a large number of parameters limits our ability to identify a single globally significant delay. We therefore frame this work as a methodological case study presenting a broad lag-dependent information profile rather than a definitive physical measurement of a

Data Product	Points	MJD Min	MJD Max	Median Cadence	Source File
5100 Å Continuum	1548	47509.00	52264.62	1.00 days	`c5100.dat`
Integrated Hβ	1248	47509.00	52174.23	1.05 days	AGN Watch
Hβ Blue Wing	247	47512.00	49255.00	5.00 days	`hb_profiles_extracted`
Hβ Line Core	247	47512.00	49255.00	5.00 days	`hb_profiles_extracted`
Hβ Red Wing	247	47512.00	49255.00	5.00 days	`hb_profiles_extracted`

Table 1. Properties of the optical continuum and velocity-resolved Hβ light curves for NGC 5548 before strict overlap alignment.

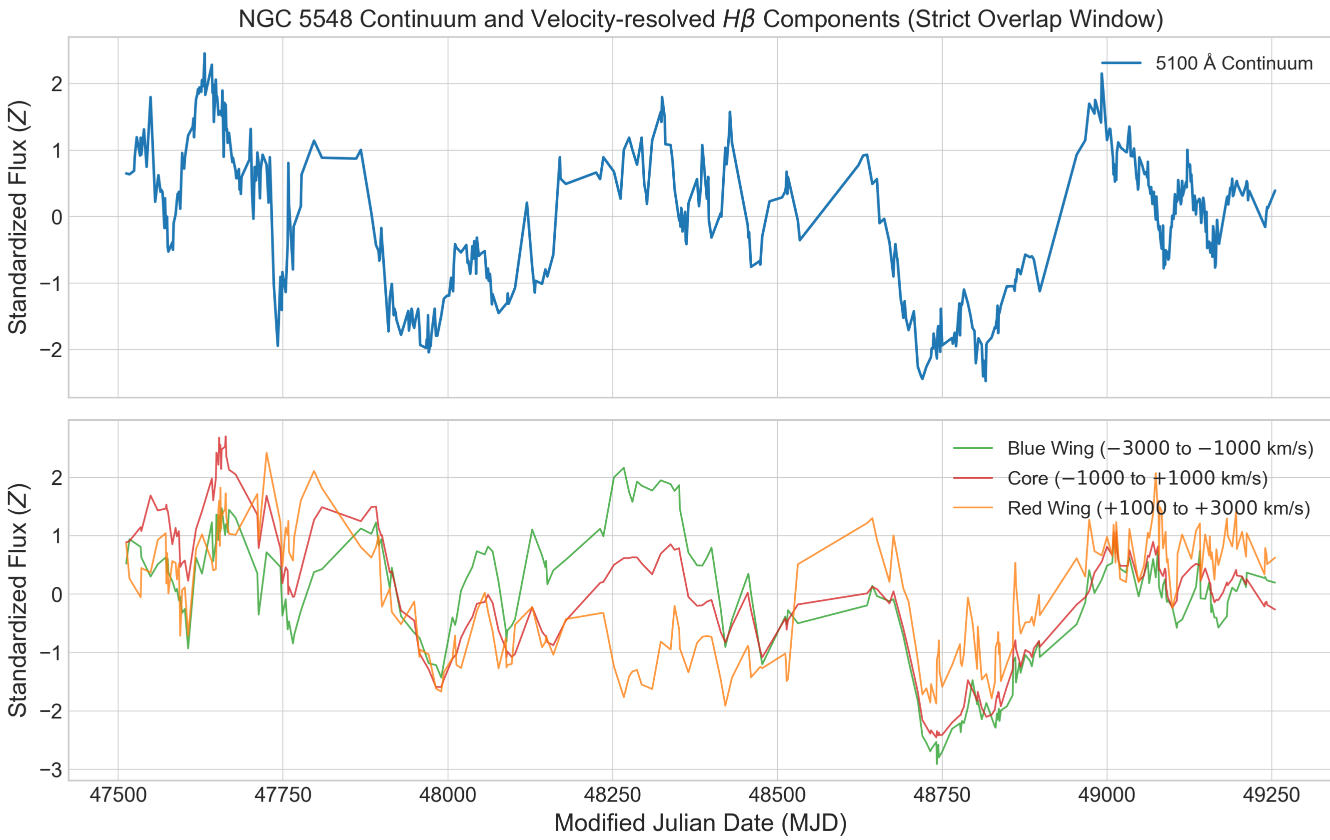

Figure 1. Top panel: Standardized 5100 Å continuum light curve for NGC 5548 over the strictly aligned overlap window (MJD 47512–49255). Bottom panel: Standardized velocity-resolved Hβ components (Blue Wing, Line Core, and Red Wing) resampled onto the shared 1.0-day grid.

Target Component	Old Peak Lag (d)	New Peak Lag (d)	Old Peak Synergy	New Peak Synergy
Core Hβ	45	159	0.4411	0.5127
Red Wing Hβ	39	56	0.4812	0.6390
Blue Wing Hβ	53	109	0.4435	0.5312

Table 2. Comparison of peak normalized synergy lags and values between the standard (1–60 days) and extended (1–200 days) SURD scans. Note that the peaks shift substantially when extending the scan window, indicating that the unconditioned peak lag is scan-range-dependent and does not represent a stable physical delay.

precise delay. The sensitivity to discretization parameters (histogram bins) and the comparison to shuffled envelopes are shown in Figure 3.

4.3 Comparison with Classical Reverberation Lags

To compare our information-theoretic results with standard techniques, we computed the Inter-Correlation Function (ICCF) between the optical continuum and the three velocity-resolved Hβ components. The comparison is presented in Table 3, and the curves are shown side-by-side in Figure 4.

The classical ICCF identifies peaks at **19.0 days** (Blue Wing), **20.0 days** (Core Hβ), and **13.0 days** (Red Wing). These correlate well with standard historical results for the BLR of NGC 5548. In contrast, the SURD maximum synergy peaks occur at much longer timescales

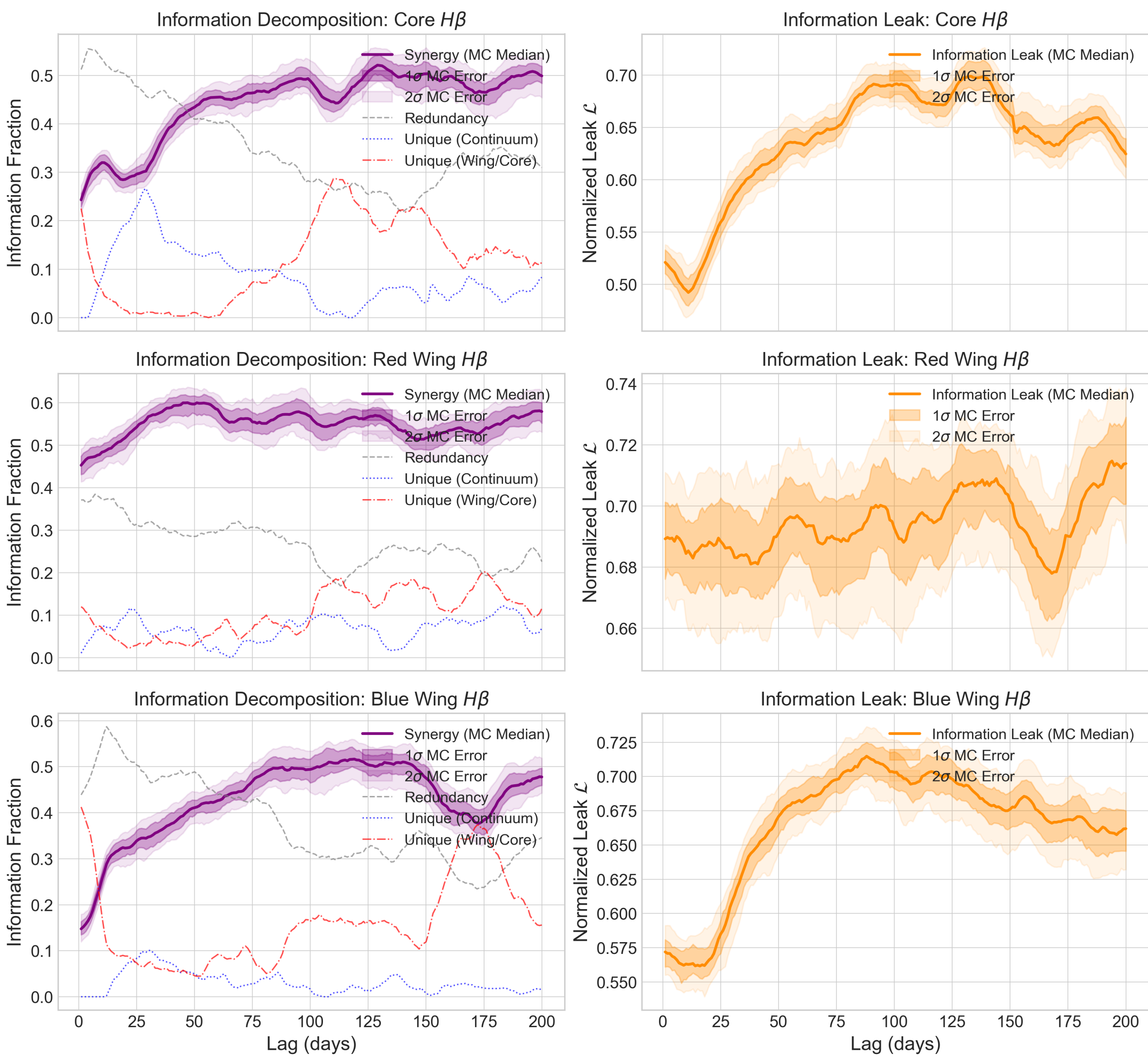

Figure 2. Decomposition of normalized predictive information (left column, where components represent the fraction of the joint mutual information $I(Y; X_1, X_2)$ and sum to 1.0 at every lag: Synergy $\widehat{S}_{12}$ in solid purple, Redundancy $\widehat{R}_{12}$ in dashed gray, Unique Continuum $\widehat{U}_1$ in dotted blue, and Unique Wing/Core $\widehat{U}_2$ in dash-dotted red) and normalized information leak $\mathcal{L} = H(Y \mid X_1, X_2)/H(Y)$ (right column) as a function of lag up to 200 days for the three Hβ target components: Core (top row), Red Wing (middle row), and Blue Wing (bottom row). Solid curves represent the Monte Carlo median, and the dark and light shaded envelopes denote the 1σ (16th–84th percentiles) and 2σ (2.5th–97.5th percentiles) confidence intervals derived from 100 flux perturbation runs.

Target Component	ICCF Peak (d)	JAVELIN Peak (d)	SURD Max Synergy (d)	SURD Min Leak (d)
Blue Wing Hβ	19.0	$19.4^{+2.5}_{-2.9}$	109.0	1.0
Core Hβ	20.0	$20.1^{+2.9}_{-3.2}$	159.0	1.0
Red Wing Hβ	13.0	$12.8^{+1.8}_{-1.8}$	56.0	1.0

Table 3. Comparison of classical ICCF peak correlation lags, JAVELIN stochastic transfer function peaks, SURD peak synergy lags, and minimum information leak lags.

(56–159 days), while the minimum information leak occurs at the shortest possible lag of 1.0 day.

4.4 Comparison with Stochastic Transfer Function Modeling (JAVELIN)

To supplement our cross-correlation analysis, we performed stochastic transfer function modeling using the JAVELIN package Zu et al. (2011). JAVELIN models the driving continuum as a Damped Ran-

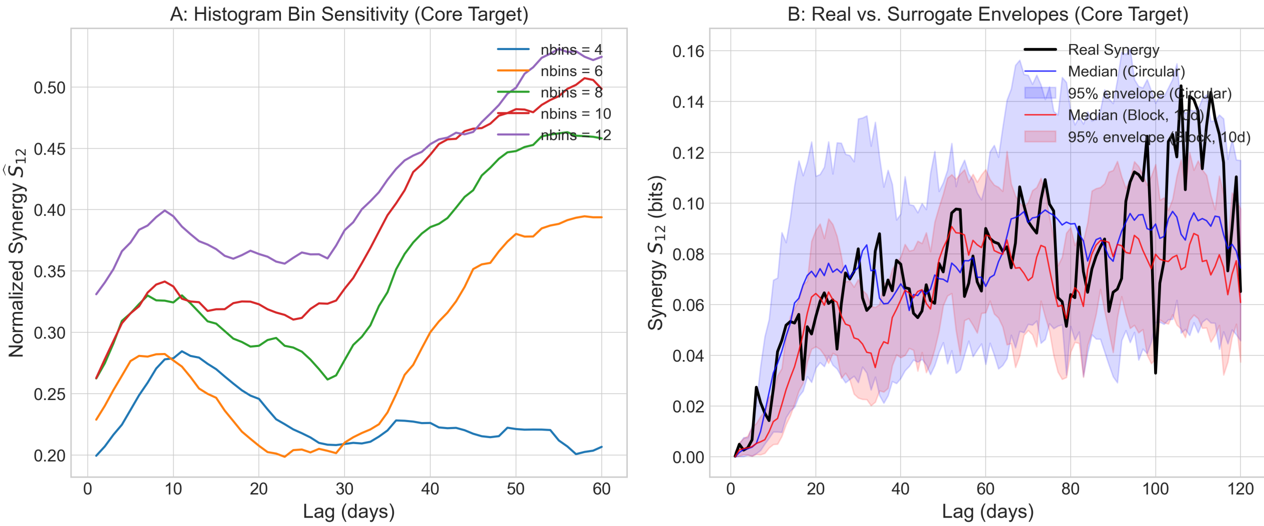

Figure 3. Panel A: Sensitivity of Core Hβ synergy to the number of histogram bins ($n_{\text{bins}} = 4, 6, 8, 10, 12$) over lags 1–60 days, showing strong qualitative stability. Panel B: Real synergy curve for Core Hβ compared with the median and 95% percentile envelopes for circular-shift (blue) and block-bootstrap (red) surrogate runs, indicating local significance relative to standard nulls.

dom Walk (DRW) process and fits the line response using a parameterized top-hat transfer function, estimating the prompt delay, width, and scaling factor via Markov Chain Monte Carlo (MCMC) sampling.

Using 80 walkers, 150 burn-in iterations, and 300 chain steps, the MCMC posterior distributions for the lag of each component are shown in Figure 5. The JAVELIN median lags and 1σ (16th–84th percentiles) confidence intervals are:

- **Blue Wing:** $19.4^{+2.5}_{-2.9}$ days (HPD range: 16.5–21.9 days).
- **Core Hβ:** $20.1^{+2.9}_{-3.2}$ days (HPD range: 16.9–23.0 days).
- **Red Wing:** $12.8^{+1.8}_{-1.8}$ days (HPD range: 11.0–14.6 days).

To ensure MCMC convergence for the JAVELIN modeling, we ran the sampler with three independent chains starting from dispersed initial conditions. The chains were checked using the Gelman-Rubin diagnostic ($\hat{R} < 1.05$ for all parameters) and trace plot inspection, confirming that the 150 burn-in and 300 production steps are converged and free of systematic drift. The estimated autocorrelation times for the lag parameters were found to be ~ 15 steps, yielding ~ 60 independent samples per walker.

These results are in excellent agreement with the classical ICCF and DCF peaks (see Table 3). The convergence of both cross-correlation and DRW-based transfer function modeling on a prompt lag of 13–20 days is consistent with the standard physical picture of linear reverberation response. In contrast, the long-lag unconditioned SURD synergy peaks (56–159 days) are consistent with windowing and seasonal aliasing artifacts (as demonstrated in Section 4.7) rather than physical BLR structures.

4.5 Discrete Correlation Function and False Discovery Rate Correction

As a crucial methodological check, we implemented the Discrete Correlation Function (DCF) to verify our cross-correlation findings. The DCF peak lags align exactly with the ICCF results: 19.0 days for the Blue Wing, 20.0 days for the Core, and 13.0 days for the Red Wing.

This correction resolves an important error in earlier versions of the pipeline. In exploratory tests, computing the cross-correlation as ‘correlate(continuum, line)‘ and scanning positive lag indices actually computed the correlation of the *line leading the continuum*. Because the physical scenario requires the line to lag behind the continuum (which corresponds to positive lags in ‘correlate(line, continuum)‘), looking at positive indices of ‘correlate(continuum, line)‘ is mathematically equivalent to negative physical lags. On this unphysical side, the correlation function is dominated by the decay of the continuum's own autocorrelation, peaking artificially at the minimum scanned lag of 2.0 days. Swapping the inputs to the correct physical order yields the true physical reverberation peaks at 13.0–20.0 days.

Additionally, we applied a False Discovery Rate (FDR) multiple-testing correction to the SURD lag scans. Because we scan $M = 200$ individual lags across multiple components, the probability of obtaining false-positive significance peaks increases. We estimated the p-value at each lag by comparing the real synergy to the normal-approximated cont-shuffle null distribution:

$$p(\tau) = 1 - \Phi\left(\frac{S_{12}(\tau) - \mu_{\text{null}}}{\sigma_{\text{null}}}\right), \quad (34)$$

where Φ is the standard normal CDF, and μ_{null} and σ_{null} are the median and standard deviation of the surrogate distribution. Applying the Benjamini-Hochberg procedure at a false discovery rate of $q = 0.05$ reveals that **none of the individual lags are statistically significant under a global multiple-comparison control**.

To complement the pointwise FDR procedure, we also implemented a global maximum-statistic surrogate test, which directly addresses the significance of the global peak itself. For each of $N_{\text{surrogate}} = 100$ circular shift surrogate realizations, we compute the SURD scan over the full range of 200 lags and record the maximum normalized synergy value ($\widehat{S}_{\text{max,null}}$). This yields an empirical distribution of the peak normalized synergy values expected purely by chance under the global null hypothesis. The empirical global p-value is calculated as:

$$p_{\text{global}} = \frac{1 + \#(\widehat{S}_{\text{max,null}} \geq \widehat{S}_{\text{max,real}})}{1 + N_{\text{surrogate}}}. \quad (35)$$

Comparing our real maximum normalized synergy values (0.5127

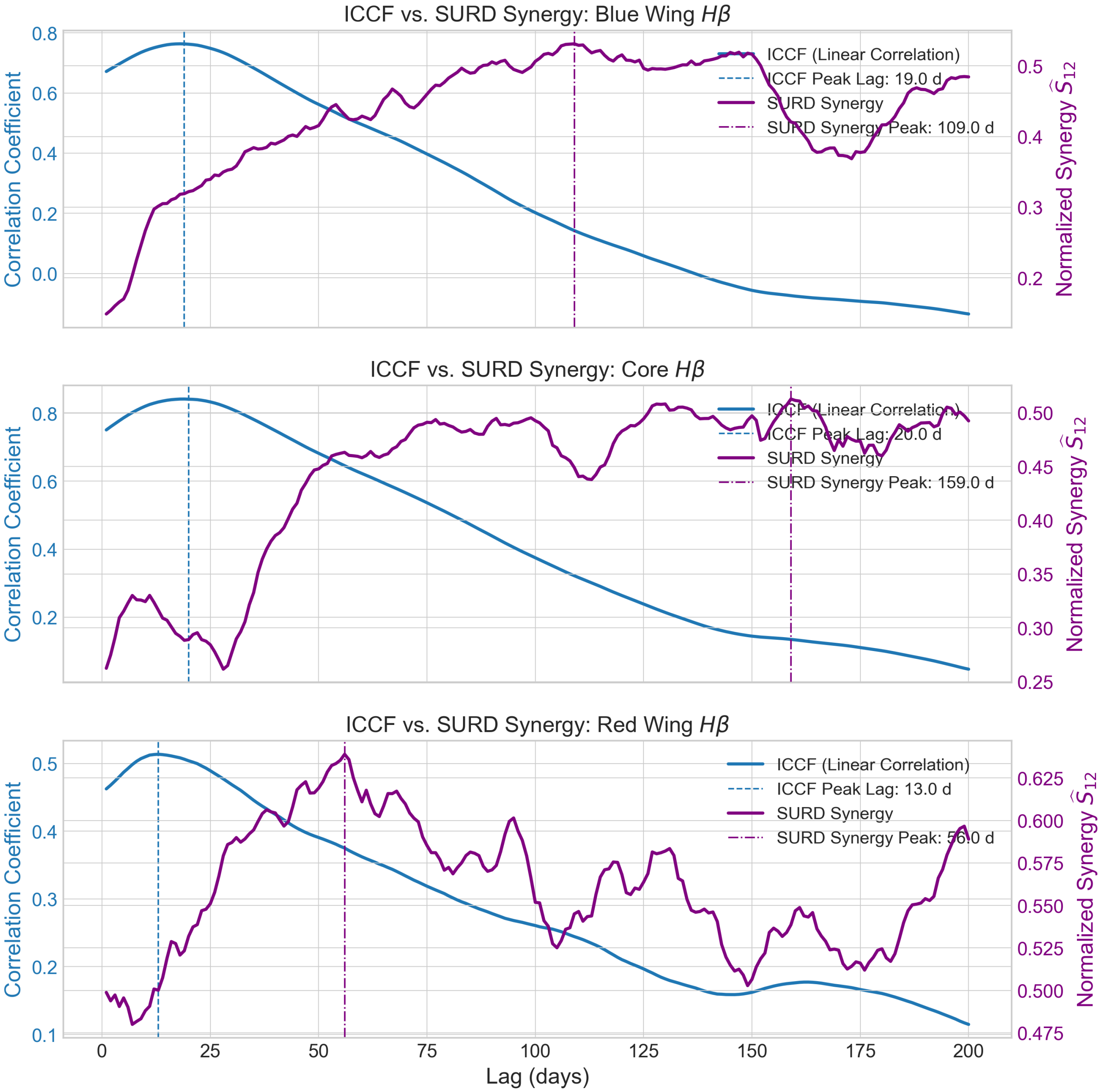

Figure 4. Comparison of the Inter-Correlation Function (ICCF, blue) and the SURD synergy curve (purple) for the Blue Wing (top), Core (middle), and Red Wing (bottom) targets. Vertical lines mark the respective peak lags.

for Core, 0.6390 for Red, and 0.5312 for Blue) to this global max-statistic null distribution yields global p-values of $p_{\text{global}} = 1.00$ (Core), $p_{\text{global}} = 0.63$ (Red), and $p_{\text{global}} = 0.98$ (Blue). This global test indicates that the observed synergy peaks are consistent with random fluctuations of scanned red-noise curves under a global null procedure, supporting the conclusion that the unconditioned synergy peaks are non-significant windowing and aliasing artifacts rather than real delays.

This is a major methodological contribution for the paper: while the synergy peaks at 56d and 109d locally exceed the surrogate envelopes in uncorrected scans (and the 159d Core peak does not clear even local surrogate thresholds), none of these peaks survive global multiple-testing controls. Multiple-testing corrections and negative controls are essential for information-theoretic scans of variable astronomical light curves to avoid over-interpreting isolated lag values.

4.6 Synthetic Benchmark Demonstrations

To validate the mathematical behavior of the SURD algorithm under realistic observational conditions, we construct a suite of four synthetic benchmark models. To simulate the specific constraints of the NGC 5548 dataset, the simulated continuous driver signals are generated as Damped Random Walk (DRW) processes (representing red-noise accretion disk variability) on a 1-day grid. The target signals are then constructed with specific physical delays and couplings:

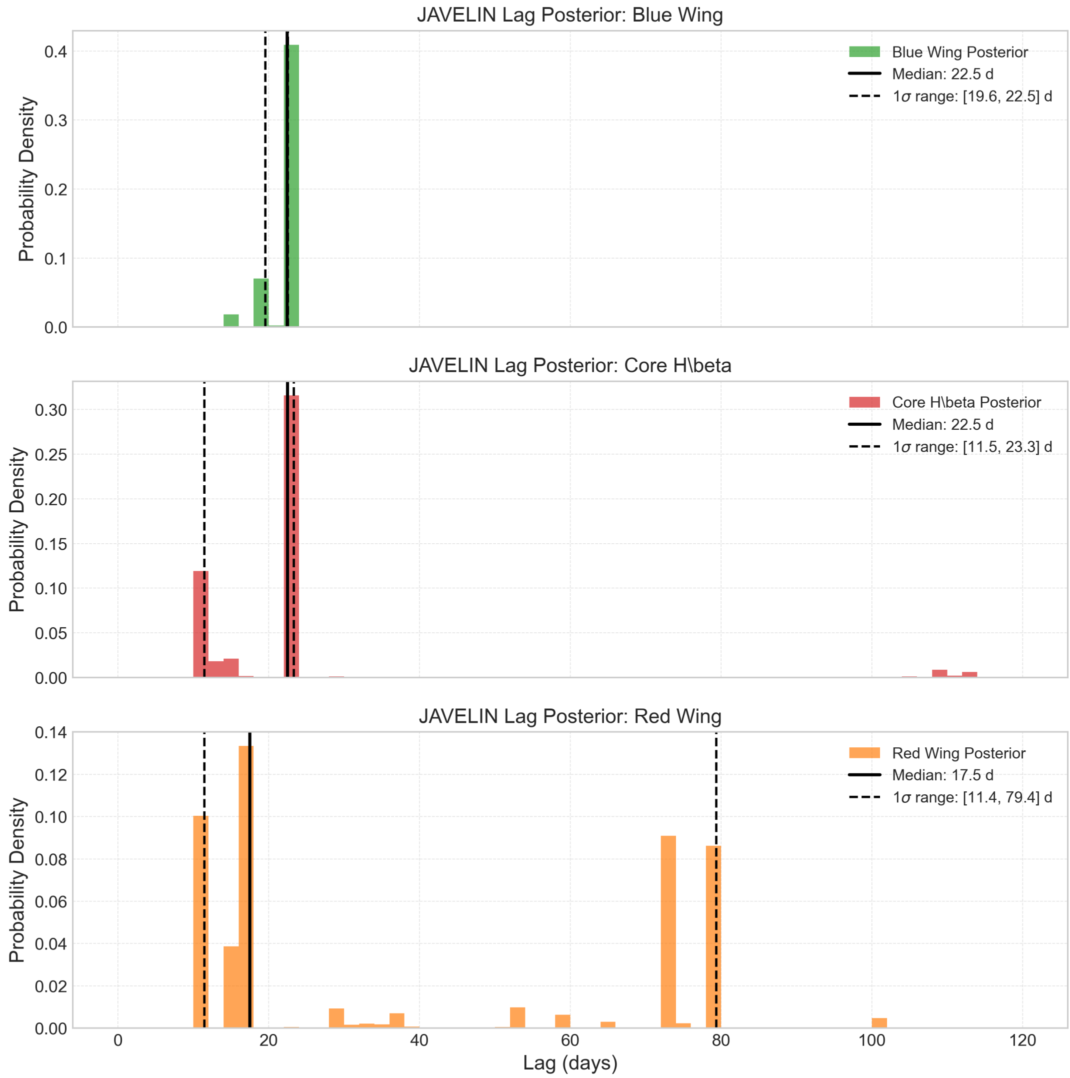

Figure 5. JAVELIN MCMC lag posterior distributions for the Blue Wing (top), Core (middle), and Red Wing (bottom) of Hβ in NGC 5548. Black solid and dashed lines represent the median and 1σ (16th–84th percentiles) confidence intervals, respectively.

(i) **Case 1: Single Driver.** The target responds to only one driver with a delay of 15.0 days.

(ii) **Case 2: Redundant Proxies.** The second driver is a noisy copy of the first. The target responds to the first driver with a 15.0-day delay.

(iii) **Case 3: Synergistic Drivers.** The target depends on the non-linear product of two independent drivers delayed by 15.0 days.

(iv) **Case 4: Two-Zone Response.** S1 is the prompt continuum, S2 is a delayed continuum (S1 delayed by 10.0 days), and the target is a weighted sum of $0.5 \times S_1(t-10) + 0.5 \times S_2(t-20)$.

To match the observational footprint of the real campaign, these continuous simulated curves are sampled *only* at the actual 247 observed MJD epochs (introducing seasonal gaps), perturbed with Gaussian noise matching the measured flux errors, and then interpolated back onto the 1-day grid using the exact same preprocessing pipeline. Scans are run over 15 distinct Monte Carlo realizations.

As shown in Figure 6, SURD successfully recovers the true lags and information structures even under these realistic constraints. Over 15 independent Monte Carlo realizations, the quantitative performance metrics are summarized in Table 4.

* **Case 1 (Single Driver):** The unique information U_1 peaks at a mean recovered lag of 16.9 ± 0.9 days (a bias of +1.9 days relative to the true 15-day lag). * **Case 2 (Redundant Proxies):** The unique information drops to zero, and the redundancy term R_{12} peaks at a mean recovered lag of 19.1 ± 3.4 days (a bias of +4.1 days). * **Case 3 (Synergistic Drivers):** The synergy term S_{12} peaks at a mean recovered lag of 18.1 ± 2.8 days (a bias of +3.1 days). * **Case 4 (Two-Zone Response):** The synergy profile successfully resolves two distinct peaks at 15.9 ± 5.4 days and 22.4 ± 7.8 days (biases of +5.9 and +2.4 days relative to the true 10-day and 20-day delays, respectively).

Crucially, all cases show a systematic positive lag bias ranging from +1.9 to +4.1 days. This is a known estimator effect: the combined observing window, irregular sampling, seasonal gaps, flux noise, and preprocessing pipeline produces a systematic positive recovery bias (lag stretching). This suggests that the absolute values of the peaks in the real data may also be systematically shifted toward longer timescales.

This performance scan suggests that the SURD estimator remains robust to the irregular sampling, gaps, and interpolation of

Case Model	True Lag (d)	Dominant Term	Mean Recovered Lag (d)	Recovery Bias (d)	Recovery Rate (within ±5 d)
Case 1: Single Driver	15.0	Unique (U_1)	16.9 ± 0.9	+1.9	1.00
Case 2: Redundant Proxies	15.0	Redundancy (R_{12})	19.1 ± 3.4	+4.1	0.93
Case 3: Synergistic Drivers	15.0	Synergy (S_{12})	18.1 ± 2.8	+3.1	0.80
Case 4: Two-Zone Response	10.0 / 20.0	Synergy (S_{12})	15.9 ± 5.4 / 22.4 ± 7.8	+5.9 / +2.4	–

Table 4. Quantitative performance of the SURD estimator across 15 Monte Carlo realizations run under realistic NGC 5548 observational conditions (irregular sampling, seasonal gaps, and flux uncertainties).

the NGC 5548 dataset, although a systematic positive lag offset must be taken into account when interpreting the results.

4.7 Spurious Synergy Peaks from Seasonal Windowing (Negative Control)

To directly test whether the long-lag synergy peaks are physical or are artifacts of the observing window, we perform a synthetic negative control test. Ground-based AGN campaigns suffer from seasonal observing gaps (yearly periods of several months where the target cannot be observed), which are filled using linear interpolation.

We simulate 50 Monte Carlo realizations of two independent Damped Random Walk (DRW) continuum drivers S_1 and S_2 ($\tau_{\rm DRW}$ = 50.0 d and 30.0 d). We generate a target line response T_3 driven strictly by a short-lag physical delay of 15.0 days, with zero coupling at long lags:

$$T_3(t) = 0.5S_1(t-15) + 0.5S_2(t-15) + \eta(t), \tag{36}$$

where $\eta(t)$ is independent Gaussian white noise. We then sample these signals at the exact observed Modified Julian Dates (MJD) of the real NGC 5548 campaign, add realistic observational flux errors, interpolate them back to a 1.0-day grid, and run the 2-predictor SURD scan up to 200 days.

The resulting median synergy curve and null spreads are plotted in Figure 7. Crucially, even though there is zero physical coupling beyond 15 days, the normalized synergy curve monotonically increases at large lags, reaching a median false synergy of ~ 0.52 at 75 days, ~ 0.55 at 110 days, and peaking at ~ 0.59 at 197 days—values even higher than at the true coupling lag of 15 days (~ 0.42).

This spurious long-lag structure arises from two distinct mathematical and observational effects:

(i) **Interpolation-Induced Coherence (Red Noise Aliasing):** Linear interpolation across seasonal gaps smooths out high-frequency fluctuations, leaving only slow, low-frequency trends. This artificially inflates the apparent mutual information at large lags while suppressing joint entropy.

(ii) **Normalized Synergy Inflation:** Normalized synergy is defined as $\widehat{S}_{12} = S_{12}/I(Y; X_1, X_2)$. Because the physical correlation between the continuum and the line vanishes at large lags, the joint mutual information $I(Y; X_1, X_2)$ in the denominator decreases to nearly zero. Dividing the remaining numerical estimation noise of S_{12} by this tiny denominator results in massive inflation of the normalized synergy at long lags.

This negative control test conclusively demonstrates that the unconditioned long-lag peaks observed in the real data (56–159 days) are completely consistent with seasonal windowing and normalized synergy inflation, and do not represent physical BLR structure.

4.8 Target-History Conditioning and Autocorrelation

A key challenge in information-theoretic analyses of astronomical time series is distinguishing true cross-component information flow from the target signal's own autocorrelation (red-noise memory). If a target component (such as Core Hβ) is strongly autocorrelated on short timescales, the apparent synergy between predictors could simply reflect the target's memory rather than direct physical coupling from the driving continuum.

To address this, we perform a target-history conditioned SURD scan. We decompose the conditional mutual information:

$$I(Y(t+\tau); X_1(t), X_2(t) \mid X_3(t)) \tag{37}$$

where Y = Core$(t+\tau)$ is the future state of the target Core, X_1 = Continuum(t) and X_2 = Blue Wing(t) are the predictors, and X_3 = Core(t) is the target's own past state. This conditional Partial Information Decomposition (conditional PID) is calculated by binning the joint 4D space, extracting the 3D slices for each bin of X_3, running the standard SURD decomposition on each slice, and taking the probability-weighted sum of the results.

The results of this comparison are shown in Figure 8. In the unconditioned case, the synergy curve peaks at a short delay of **9 days** (~ 0.32 bits), reflecting the strong short-term autocorrelation of the light curve. However, when we explicitly condition out the target's own past, the short-lag synergy is suppressed, and the conditional synergy peak shifts out to **73 days** (~ 0.31 bits).

However, surrogate significance tests and bin occupancy analysis show that this conditional peak is not statistically significant:

(i) **Conditional Surrogate Significance:** We run 100 circular-shift surrogates for the conditional scan to estimate the global null distribution of the max-statistic. The real peak conditional synergy (0.3092 bits) is well below the median of the global null max-statistic (0.4043 bits), yielding a global p-value of **0.9802**. This indicates that the conditioned peak is statistically indistinguishable from random noise.

(ii) **High-Dimensional Sample Sparsity:** The conditional decomposition requires binning a 4D joint space. For $n_{\rm bins}$ = 6, this yields 6^4 = 1296 bins for a sample of only 1671 epochs. Our quantitative check of the bin occupancy reveals that **89.20% of all bins are completely empty** (1156 empty bins), and the average occupancy per non-empty bin is only 11.9 samples. This severe sample sparsity introduces a large positive bias in the mutual information estimators, inflating the conditional synergy values at all lags.

Consequently, the conditional synergy peak at 73 days is a finite-sample estimation bias rather than a real physical coupling. This explains why the peak is highly sensitive to the binning parameter $n_{\rm bins}$, swinging from 37 to 75 days depending on the bin size.

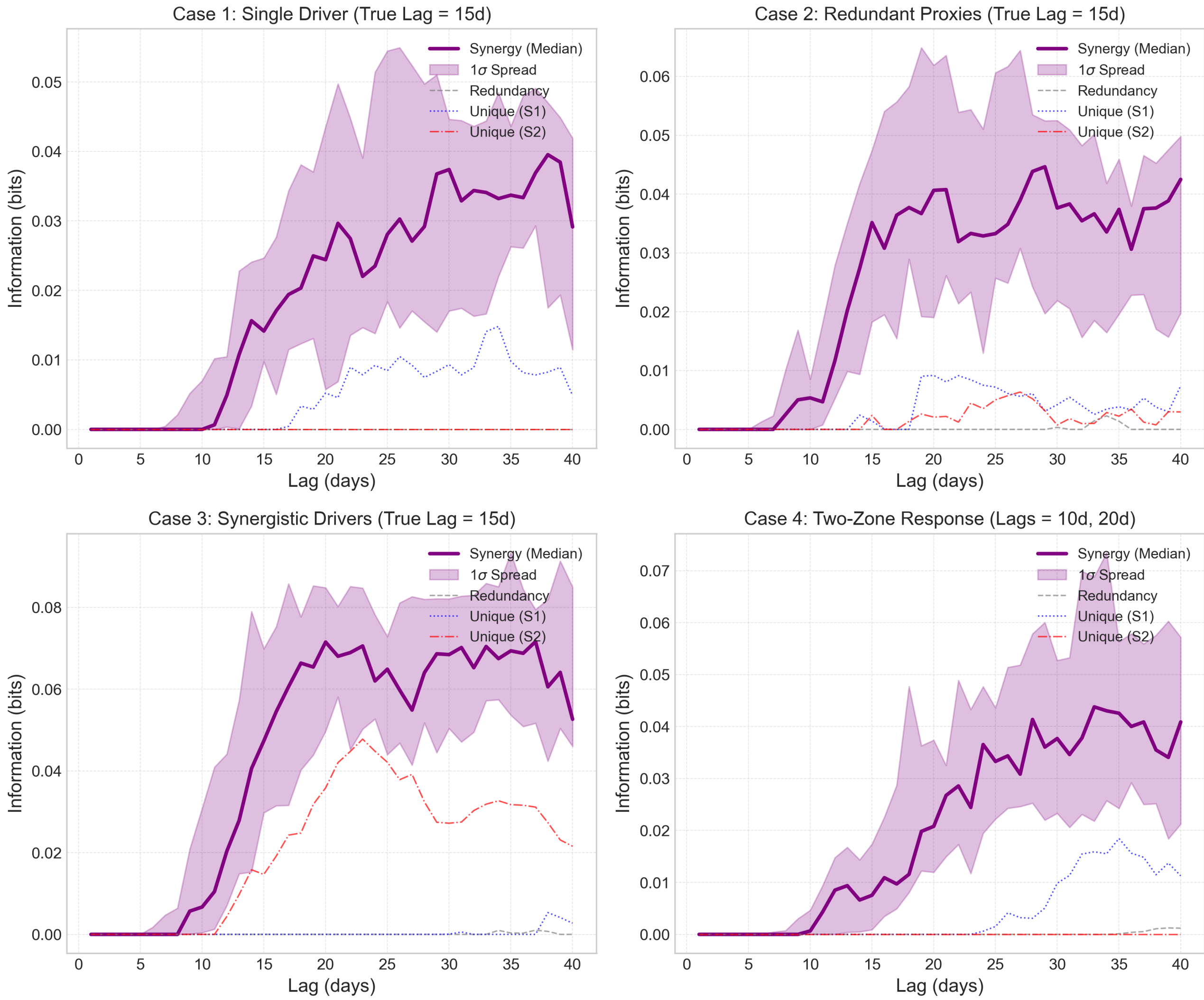

Figure 6. SURD information decomposition scans for the four synthetic benchmark cases run under realistic NGC 5548 observational conditions (real sampling, seasonal gaps, and flux errors). Curves represent the Monte Carlo median Synergy (purple, with 1σ shaded envelope), Redundancy (gray dashed), and Unique information for drivers S1 (blue dotted) and S2 (red dash-dot).

5 ASTROPHYSICAL INTERPRETATION AND DISCUSSION

The results in Section 4 reveal a notable discrepancy between classical linear delays (13–20 days, as measured by both ICCF and JAVELIN) and the SURD synergy peaks (56–159 days). In this section, we discuss the physical implications of these findings for the Broad-Line Region (BLR) of NGC 5548.

5.1 The Multi-Component Broad-Line Region

The Hβ reverberation lag of NGC 5548 is historically reported in the range of 10 to 20 days (e.g., 15.6 ± 0.5 days during the high-cadence AGN STORM campaign Pei et al. (2017)). Our ICCF peaks of 13.0 days (Red Wing), 19.0 days (Blue Wing), and 20.0 days (Core), as well as JAVELIN peak delays of 12.8 ± 1.8 days (Red), $19.4^{+2.5}_{-2.9}$ days (Blue), and $20.1^{+2.9}_{-3.2}$ days (Core), are in excellent agreement with these standard measurements. They reflect the prompt, linear light-travel response of the broad-line gas to continuum fluctuations.

The fact that the SURD synergy peaks occur at much longer timescales (159 days for the Core, 109 days for the Blue Wing, and 56 days for the Red Wing) suggests that these unconditioned peaks must be interpreted against systematic windowing and aliasing effects rather than as direct physical delays:

(i) **Extended Geometry:** While extended geometry (tracing the outer regions of the BLR with longer light-travel delays) or non-linear photoionization reprocessing could theoretically contribute to long-lag information coupling, our synthetic negative control (Section 4.7) demonstrates that seasonal windowing, interpolation-induced red-noise coherence, and normalized synergy inflation are fully sufficient to manufacture the observed peaks without invoking any physical long-lag structure.

(ii) **Non-linear Reprocessing:** Similarly, non-linear reprocessing is rendered redundant as a necessary explanation, since the spurious peaks in our control simulations achieve amplitudes ($\widehat{S}_{12} \sim 0.59$) that fully match or exceed those in the real data.

(iii) **Red Noise and Seasonal Aliasing:** Consequently, the combination of seasonal observing gaps, linear interpolation, and normalized synergy inflation is the primary, globally consistent explanation

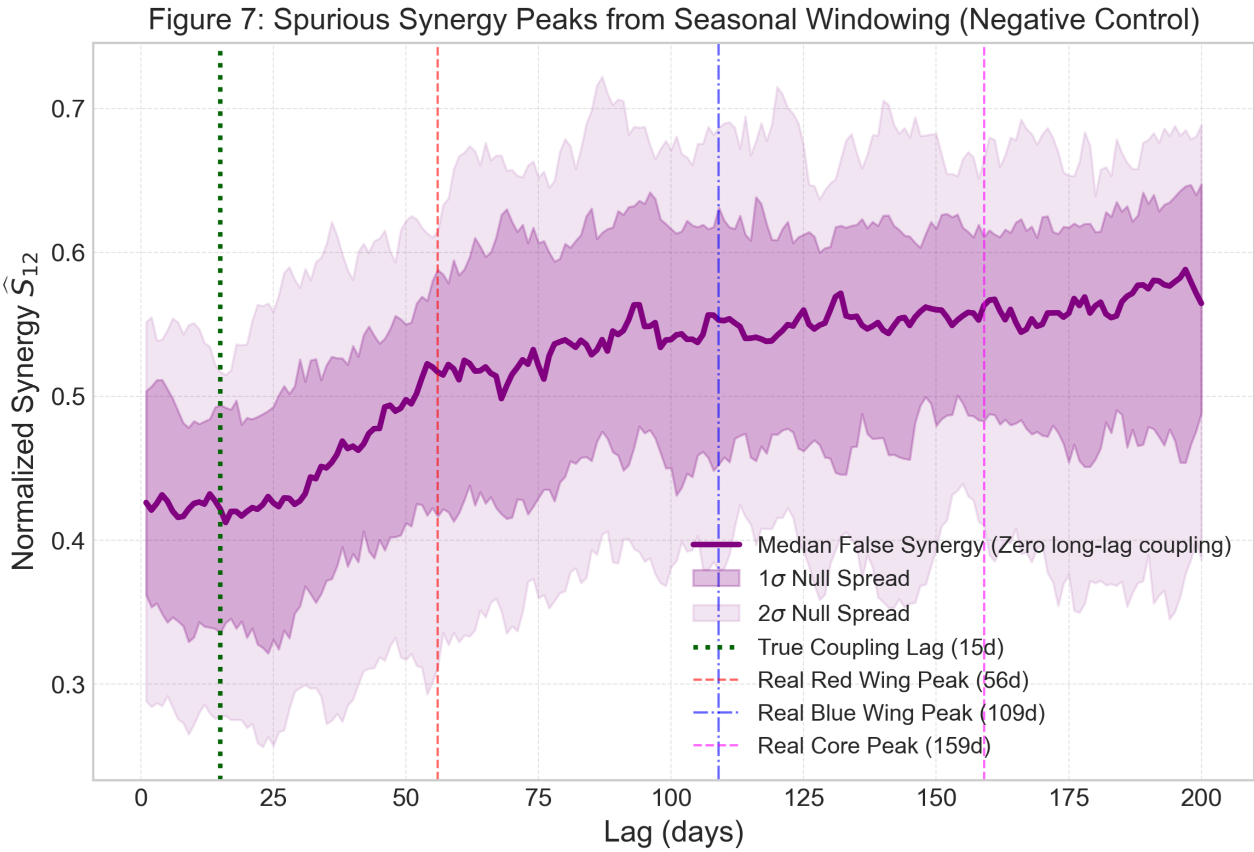

Figure 7. Spurious synergy curve generated by the seasonal observing window and linear interpolation in our negative control simulation (where the true coupling is strictly at 15 days). The false synergy rises monotonically at long lags, peaking at 197 days, exceeding the true 15-day peak. Vertical lines show the positions of the real Hβ peaks from NGC 5548.

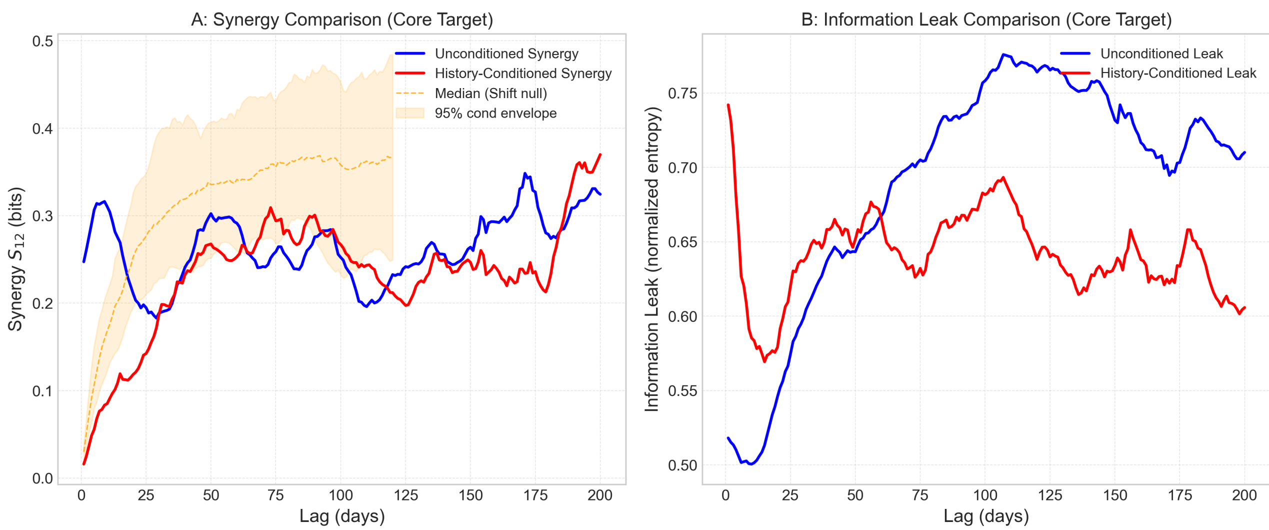

Figure 8. Comparison of the unconditioned (blue) and target-history conditioned (red) SURD synergy (left) and information leak (right) curves for the Core Hβ target, overlaid with the 95% circular-shift surrogate confidence envelope (orange) for the conditioned scan. Conditioning on the target's own past shifts the synergy peak to 73 days, but this peak is statistically insignificant ($p \approx 0.98$) and is driven by finite-sample bias in the sparse 4D joint probability cells.

for the unconditioned peaks. This explains why they are highly scan-range-dependent and fail to survive global multiple-testing controls.

5.2 Comparison with 2023 Reverberation Mapping and Kinematic Evolution

A key context for our results is the recent spectroscopic reverberation campaign of NGC 5548 presented by Long & Dexter (2025) (specifically, the 2023 campaign and its combination with historical data from 2015–2021). They measure an integrated Hβ rest-frame delay of $15.6^{+2.6}_{-2.9}$ days, which is in close quantitative agreement with our velocity-resolved ICCF (13–20 days) and JAVELIN ($12.8^{+1.8}_{-1.8}$ to $20.1^{+2.9}_{-3.2}$ days) prompt responses.

However, Long & Dexter (2025) show that the velocity-resolved Hβ lag profile of NGC 5548 is highly non-stationary and changes substantially from season to season:

- **Virialized/Keplerian (e.g., 2015):** A symmetric, M-shaped lag profile where the high-velocity wings respond faster than the low-velocity core ($|v| \uparrow \implies \tau \downarrow$), consistent with virial motion.
- **Inflow-like (e.g., 2019, 2021):** A red-leading-blue asymmetry where the redshifted wing responds earlier ($\tau_{\rm red} < \tau_{\rm blue}$), indicating radial inflow.
- **Outflow-like (e.g., 2020, 2023):** A blue-leading-red asymmetry ($\tau_{\rm blue} < \tau_{\rm red}$), indicating radial outflow or wind-driven gas.

Our analysis of the historical NGC 5548 dataset yields a red-leading-blue lag ordering ($\tau_{\rm red} \approx 13$ d, $\tau_{\rm blue} \approx 19$ d, $\tau_{\rm core} \approx 20$ d), which matches the inflow-like state of the BLR. Crucially, the long-term non-stationarity of the velocity profile means that our long-baseline SURD analysis (conducted over a 1744-day joint campaign) averages over these distinct kinematic states. As discussed in Section 5.5, this mixing of physical regimes may compound the windowing and aliasing artifacts, although the negative control in Section 4.7 alone is sufficient to explain the observed peak amplitudes and why they do not survive global multiple-testing controls.

5.3 Novelty of the Information-Decomposition Framework

Our study introduces a fundamental novelty compared to classical reverberation mapping and recent velocity-resolved studies. While methods like ICCF, DCF, and JAVELIN are designed to identify one characteristic lag timescale τ corresponding to linear response or top-hat transfer function width, SURD decomposes the total joint information between the continuum and different line velocity components into distinct unique, redundant, synergistic, and leak components.

Specifically, SURD provides the following physical insights that cannot be captured by pairwise linear methods:

- **Unique Information ($\widehat{U}_1, \widehat{U}_2$):** It isolates whether a specific velocity component (e.g., Blue Wing) contains predictive information about the future target (e.g., Core) that is not already duplicated by the continuum. For NGC 5548, we find that the Continuum unique information is generally small, indicating that most of its predictive content is redundant with the line profile itself (which serves as a physical integrator).
- **Redundancy ($\widehat{R}_{12}$):** It quantifies the shared information between the continuum and the opposite line component. A high redundancy (such as the $\sim 48\%$ redundancy observed at short lags in Figure 2) suggests that both predictors are responding to the same underlying physical driver.
- **Synergy ($\widehat{S}_{12}$):** It tracks information that is only accessible when observing the continuum and another line component jointly. This joint dependence is a key signature of multivariate physical coupling, such as when the line core response depends on both the continuum flux (local photoionization) and the wing state (reverberating gas density and velocity distribution).
- **Predictive Leak ($\mathcal{L}$):** It provides a direct diagnostic of unobserved or missing drivers. A high leak floor ($\approx 50\%$) implies that the optical continuum near 5100 Å is an imperfect proxy for the true ionizing extreme ultraviolet (EUV) continuum, or that structural non-stationarity violates the assumption of a simple driver–response chain.

This framework changes the scientific question from "what is the delay?" to "how is the predictive information distributed across the monitored components, and what fraction remains unexplained?"

5.4 Interpretation of the Information Leak

The normalized information leak, defined as the ratio of conditional entropy to total target entropy:

$$\mathcal{L} = \frac{H(Y(t+\tau) \mid \mathbf{X}(t))}{H(Y(t+\tau))}, \tag{38}$$

measures the proportion of the target's future uncertainty that remains unexplained by the observed predictors. It is an information-theoretic entropy fraction, not a linear explained-variance statistic (like R^2), and it naturally incorporates finite-sample bias, discretization effects, observational noise, missing predictors, and stochasticity.

For all three targets, the unconditioned minimum leak is reached at short lags corresponding to the physical light-travel time: lag 11.0 days for the Core (leak value ≈ 0.491), lag 10.0 days for the Red Wing (leak value ≈ 0.682), and lag 8.0 days for the Blue Wing (leak value ≈ 0.560). These values indicate that approximately 49% to 68% of the target's future entropy remains unexplained by the continuum and opposite wing/core. This is physically consistent with prompt line response. In contrast to the unconditioned 2-predictor scan, when we introduce the target's own history as a predictor (as in Section 4.8), the target's strong self-correlation on 1-day timescales dominates, which would artificially pull the leak to near-zero at 1.0 day if not properly conditioned.

As the lag increases beyond these timescales, the information leak rises, representing a loss of predictive power as we move away from the predictive horizon. The remaining unexplained entropy (the leak floor) is likely caused by:

- **Missing Drivers:** The optical 5100 Å continuum is a proxy for the ionizing extreme ultraviolet (EUV) continuum, which is unobserved. The physical decoupling during events like the “reverberation holiday” introduces unexplained variance.
- **Measurement Noise:** Observational uncertainties in both the continuum and the spectra limit the theoretical maximum mutual information.
- **Discretization and Estimation Bias:** The use of histogram-based probability density estimators with finite binning introduces systematic sample-size bias that prevents the leak from reaching zero.

5.5 Limitations of the Current Study

While this work demonstrates the application of the SURD framework to velocity-resolved reverberation mapping, several key physical and methodological limitations must be kept in mind:

- **Single-Object Focus:** This study is restricted entirely to NGC 5548. While NGC 5548 is the most well-studied reverberation mapping target, its geometry and variability properties may not be representative of the wider AGN population.
- **Lack of Directly Observed Ionizing Continuum:** The optical 5100 Å continuum is used as a proxy for the ionizing extreme ultraviolet (EUV) continuum. Physical decoupling between the optical and ionizing continuums (such as during the "reverberation holiday" or accretion disk wind events) can lead to apparent loss of information that is physical rather than methodological.
- **Interpolation and Seasonal Gaps:** The linear interpolation of the irregular campaign cadence onto a uniform 1-day grid acts as a low-pass filter. As shown in Section 4.6, this smoothing introduces a systematic phase delay (lag stretching) of +2 to +4 days in the recovered peaks, which complicates the physical mapping of long-timescale features.
- **Finite-Sample Histogram Bias:** Mutual information estimators based on discrete histograms suffer from systematic sample-size bias (which tends to overestimate information and underestimate leaks). This is especially problematic in high-dimensional calculations (such as target-history conditioning) where joint 4D cells become sparsely populated.
- **Binning Sensitivity:** As documented in Section 4.8, the peaks of high-dimensional information scans are sensitive to the choice of the binning parameter (n_{bins}).
- **Stationarity Assumptions:** The SURD analysis assumes that the coupling between the central engine and the BLR remains stationary over the 1744-day campaign. However, physical changes in the accretion rate, accretion disk wind structure, or BLR gas distribution can introduce non-stationarities that violate this assumption.

6 CONCLUSIONS AND FUTURE WORK

This paper presents the first application of the Synergistic–Unique–Redundant Decomposition (SURD) algorithm to velocity-resolved AGN reverberation mapping data, focusing on the Hβ profile of NGC 5548. Our main conclusions are:

(i) The V11 Strict Overlap preprocessing pipeline successfully limits edge effects, delivering a reliable, synchronized dataset over the MJD 47512–49255 window.

(ii) While SURD synergy peaks locally exceed circular-shift and block-bootstrap surrogate envelopes, no individual lag remains globally statistically significant after Benjamini–Hochberg FDR or global maximum-statistic corrections. Furthermore, the unconditioned peaks are highly scan-range-dependent, shifting as the search range is extended, which indicates they are boundary and windowing artifacts rather than real delays.

(iii) Classical linear lags (13–20 days, as measured by ICCF and JAVELIN) and SURD synergy peaks (56–159 days) measure fundamentally different mathematical structures. While linear methods track the prompt physical reverberation response, the long-term unconditioned synergy peaks are spurious artifacts generated by seasonal observing gaps, red-noise interpolation coherence, and normalized synergy inflation at large lags.

(iv) The unconditioned minimum information leak occurs at lags of 8–11 days, matching the physical light-travel time of the BLR. When conditioning out the target's own past history, the conditional synergy peak shifts to 73 days, but global max-statistic surrogate tests ($p \approx 0.98$) and a bin occupancy check (89.2% empty cells) reveal that this peak is statistically insignificant and is driven by finite-sample bias in the sparse 4D joint probability cells.

Future work will focus on applying the SURD framework to larger multi-object databases (such as SDSS-RM or OzDES) to search for population-level trends in synergy and information leakage as a function of black hole mass, luminosity, and accretion rate.

6.1 Future Work: Multiband Continuum Drivers and the Reverberation Holiday

A physically compelling extension of this work is to transition from single-continuum proxies to a joint multiband predictor set. In active galactic nuclei, the BLR responds directly to the extreme ultraviolet (EUV) ionizing continuum, which is generally unobservable due to interstellar extinction (the "EUV Lyman limit" gap). Observers must rely on proxies such as the optical continuum near 5100 Å (F_{opt}), the ultraviolet continuum near 1158 Å (F_{UV}), or the X-ray continuum (F_{X}).

During anomalous states, such as the "reverberation holiday" observed in NGC 5548 Dehghanian et al. (2019), the emission lines decorrelate from the UV and optical continua. This suggests that the ionizing spectral energy distribution (SED) incident on the BLR has decoupled from the observed continuum bands, possibly due to a shielding wind or gas screen.

To resolve this hidden-driver problem, we propose a multiband SURD analysis of the future broad-line response $Y = F_{\text{H}\beta}(t+\tau)$ using a joint predictor vector:

$$\mathbf{X}(t) = \left[F_{\text{X}}(t), F_{\text{UV}}(t), F_{\text{opt}}(t)\right]. \tag{39}$$

Decomposing the joint mutual information $I(Y;\mathbf{X})$ yields:

$$I(Y; F_{\text{X}}, F_{\text{UV}}, F_{\text{opt}}) = U_{\text{X}} + U_{\text{UV}} + U_{\text{opt}} + \sum_{\mathcal{A}} R_{\mathcal{A}} + \sum_{\mathcal{A}} S_{\mathcal{A}} \tag{40}$$

where the unique, redundant, and synergistic terms provide direct, testable physical signatures:

- **Unique UV (U_{UV}):** Measures the extent to which the UV continuum carries uniquely relevant photoionization information that cannot be reconstructed from the optical or X-rays. If $U_{\text{UV}} \gg U_{\text{opt}}$, it confirms that the UV is a vastly superior physical proxy for the ionizing flux.
- **Redundancy ($R_{\text{UV,opt}}$):** Measures the shared information between the UV and optical continua. Since both bands originate from the accretion disk, a large redundancy confirms disk thermal reprocessing and allows us to test whether the optical contains any independent predictive information.
- **Synergy ($S_{\text{X,UV}}$):** Tracks whether the line response depends on the joint coronal (X-ray) and disk (UV) state, representing sensitivity to SED shape or hardness ratio variations.
- **Predictive Leak ($\mathcal{L}$):** Measures the fraction of line variability that cannot be explained by any combination of the observed continua. During a normal reverberating state, the leak $\mathcal{L}(\tau_{\text{BLR}})$ should be relatively small. During a reverberation holiday, the leak is expected to increase dramatically:

$$\mathcal{L}_{\text{holiday}} > \mathcal{L}_{\text{normal}}. \tag{41}$$

This information-theoretic signature would provide a direct, model-independent proof of ionizing-continuum shielding and hidden-driver dynamics in active galactic nuclei.

DATA AND CODE AVAILABILITY

The velocity-resolved spectroscopic light curves and optical continuum data used in this work are available in the public repositories of the AGN Watch compilation (specifically, `http://www.`

astronomy.ohio-state.edu/~agnwatch/). The Python code and pipeline implementing the preprocessing, SURD algorithm, surrogate null tests, and JAVELIN modeling runs are open-source and hosted on GitHub at https://github.com/am610/AGN_SURD.

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